

Patient Phenotyping via Full-Dataset Immunological & Genomic Clustering

We performed unsupervised subtyping across the **entire combined study dataset** ($N = 699$ patients across **TCGA-SKCM**, **Liu 2019**, **Hugo 2016**, and **Riaz 2017**) using Agglomerative Hierarchical Clustering (Ward linkage). To prevent technical study platform offsets from dominating the clustering, feature scores (immune signatures, TMB, and age) were Z-score standardized **within each cohort** prior to pooling.

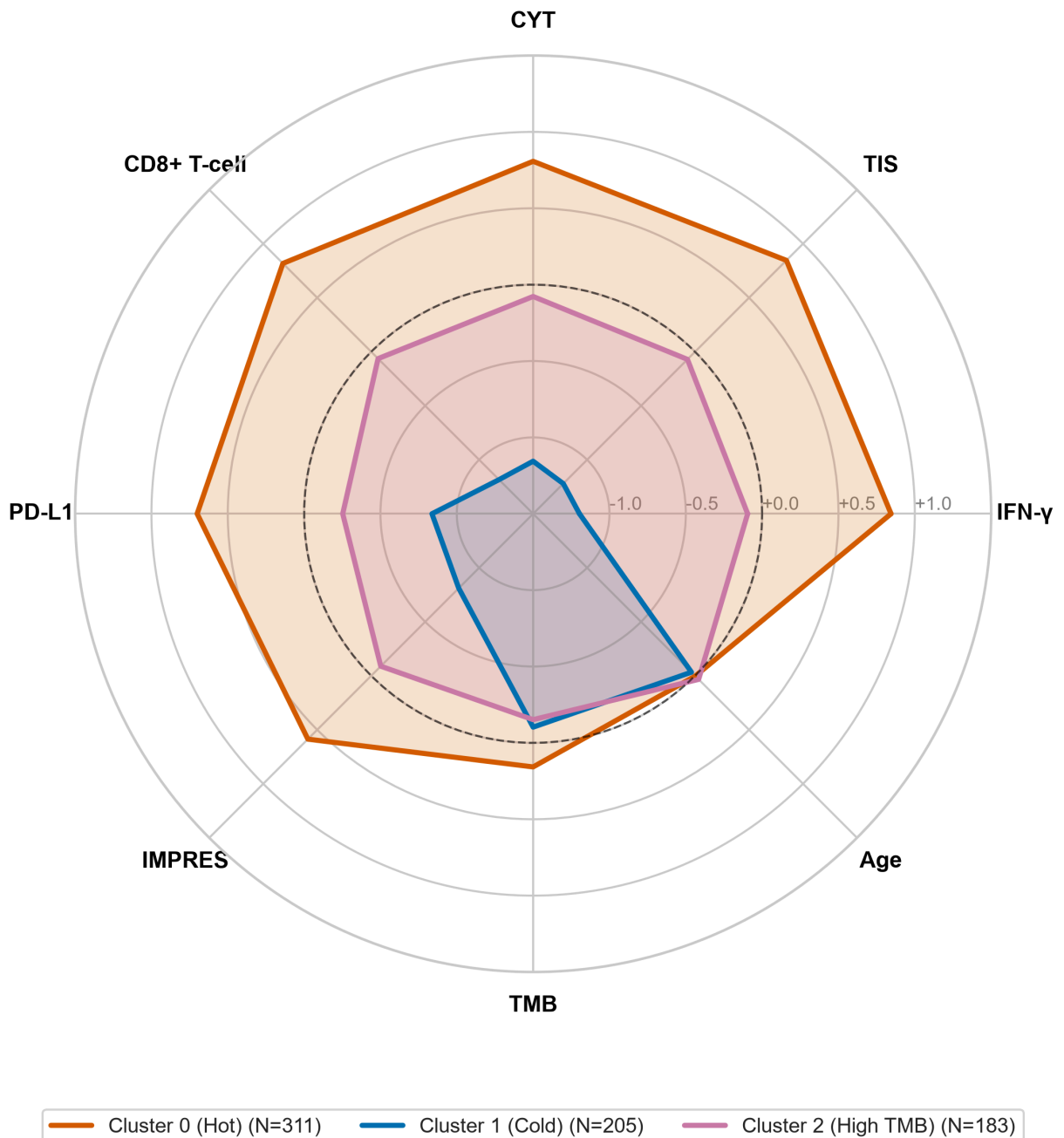
Subtype Profiles

Unsupervised subtyping isolates three distinct patient phenotypes across the $N = 699$ population. Multi-dimensional Z-score signature profiles and annotated feature heatmaps for each patient subtype are visualised below:

1. Multi-Dimensional Phenotype Fingerprint (Radar Profile)

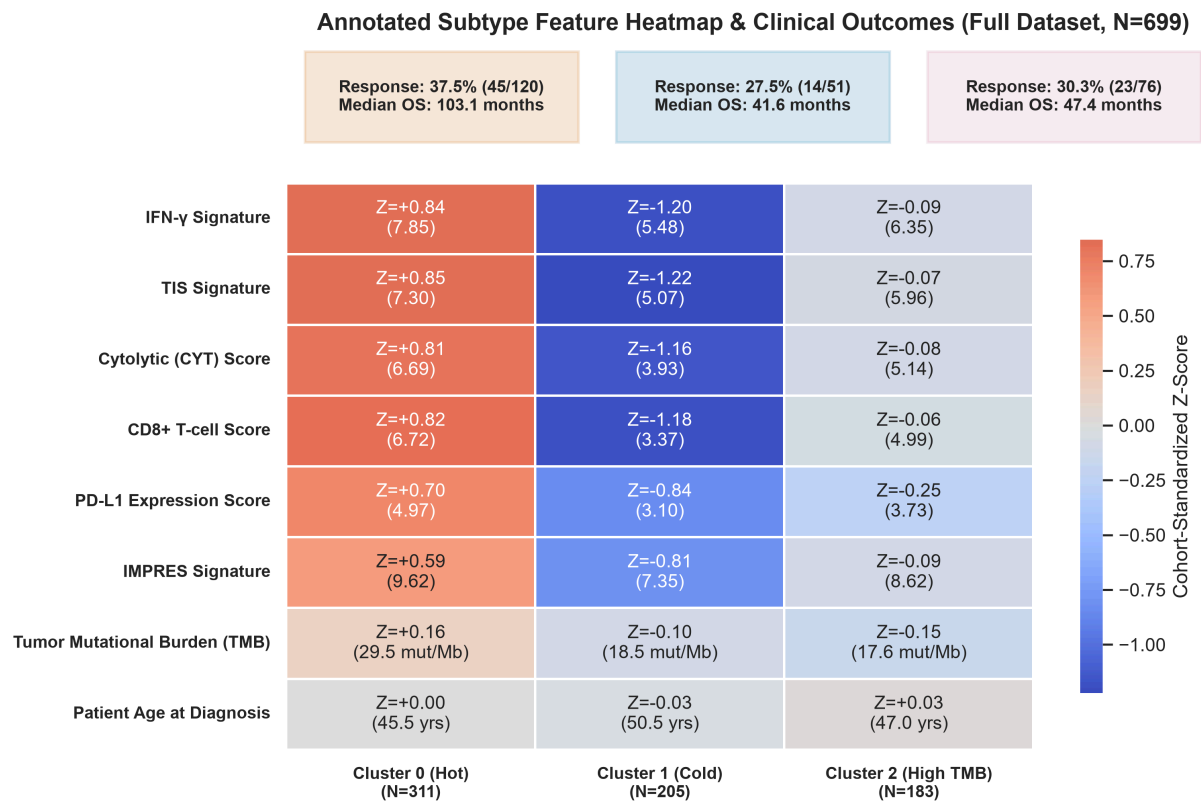
The polar radar chart below displays the standardized Z-score profiles across transcriptomic immune signatures, mutational burden, and patient age for each subtype:

Multi-Dimensional Phenotype Fingerprint by Patient Subtype (N=699)



2. Annotated Subtype Feature Heatmap & Clinical Tracks

The heatmap below details the Z-score signature matrix for each patient cluster, annotated with objective immunotherapy response rates (CR/PR %) and median overall survival (OS):



Key Analytical Findings

- Prognostic Stratification** ($N = 699$): Hierarchical clustering on within-cohort Z-score standardized features yields a highly statistically significant overall survival separation across the full 4-cohort dataset (Log-Rank $p = 2.29e - 05$). Patients in the **Immunologically Hot** cluster achieve a median survival of **103.1 months**, substantially longer than the **Cold** cluster (41.6 months).
- Therapeutic Response Alignment** ($N = 247$): Patients in **Cluster 0 (Hot)** demonstrate the highest objective response rate to anti-PD-1 immunotherapy (**45 (37.5%)**), compared to **14 (27.5%)** in **Cluster 1 (Cold)**, validating that unsupervised microenvironment subtyping captures anti-tumor immune responsiveness.
- Genomic vs. Transcriptomic Decoupling**: High tumor mutational burden alone (**Cluster 2**, mean TMB = 17.6 mut/Mb) yields only an intermediate overall survival trajectory (47.4 months) in the absence of robust T-cell inflammation, demonstrating that high TMB is insufficient without an active immune microenvironment.

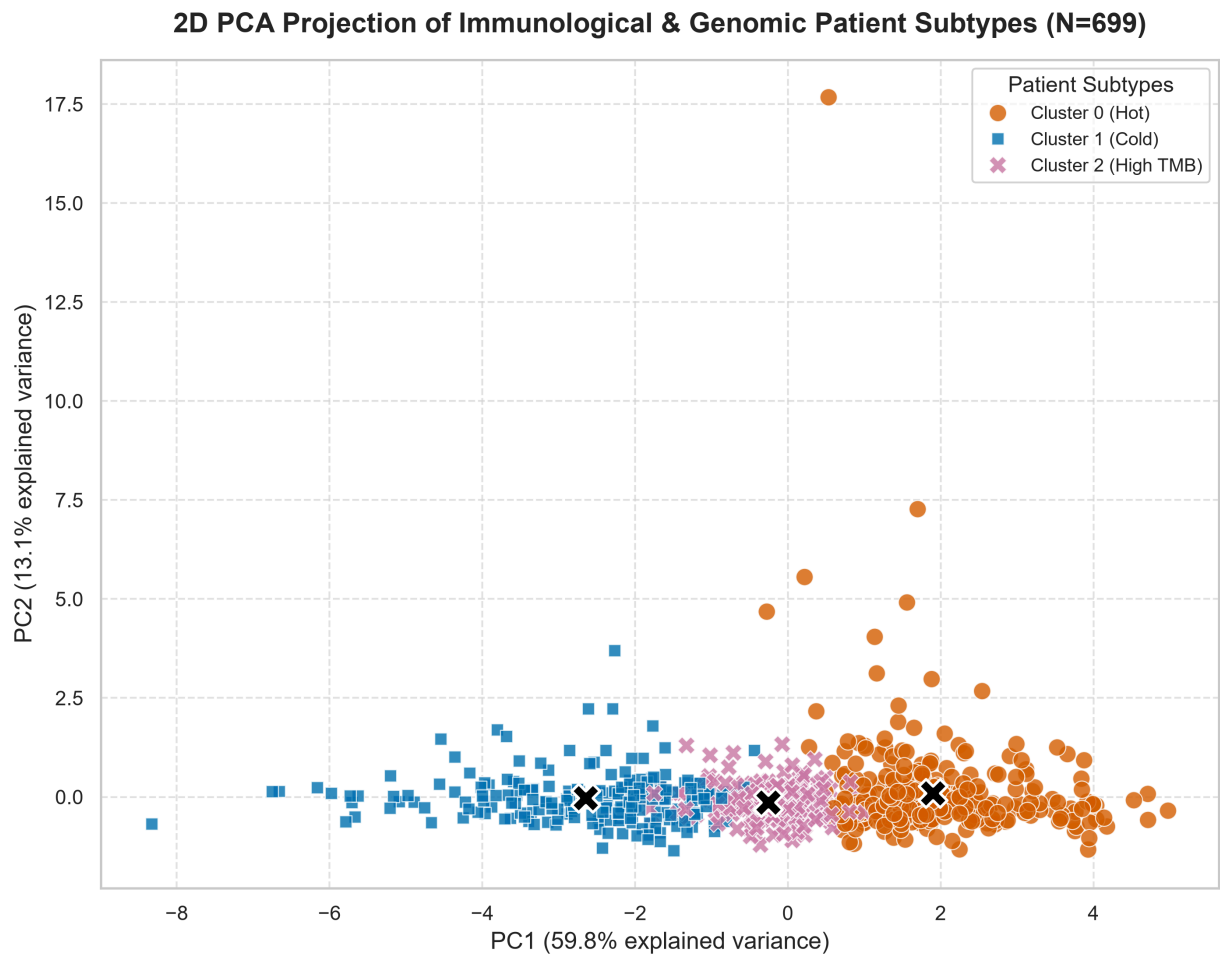
Biological Interpretation of Patient Subtypes

The unsupervised clustering isolates three distinct patient phenotypes across the multi-study population:

- Cluster 0: Immunologically Hot** ($N = 311$)
 - Immune Signatures**: Highest T-cell inflammation (IFN- γ) = 7.85, TIS = 7.30, CYT = 6.69, CD8 = 6.72).
 - Therapeutic Benefit**: Highest immunotherapy response rate (**45 (37.5%)**).
 - Prognosis**: Superior overall survival trajectory (Median OS = **103.1 months**).
- Cluster 1: Immunologically Cold** ($N = 205$)
 - Immune Signatures**: Attenuated T-cell inflammation across all markers (IFN- γ) = 5.48, TIS = 5.07, CYT = 3.93, CD8 = 3.37).
 - Therapeutic Benefit**: Lower response rate to anti-PD-1 therapy (**14 (27.5%)**).
 - Prognosis**: Poor overall survival trajectory (Median OS = **41.6 months**).
- Cluster 2: High-TMB / Hypermutated** ($N = 183$)
 - Genomics**: Highest tumor mutational burden (**TMB = 17.6 mut/Mb**).
 - Immune Signatures**: Moderate T-cell inflammation (IFN- γ) = 6.35, TIS = 5.96).
 - Prognosis**: Intermediate survival trajectory (Median OS = **47.4 months**).

Subtype Visualisation (2D PCA Projection)

Below is a 2D PCA projection showing clear multi-dimensional separation of the patient subtypes across the $N = 699$ full dataset. The 'X' markers denote cluster centroids:

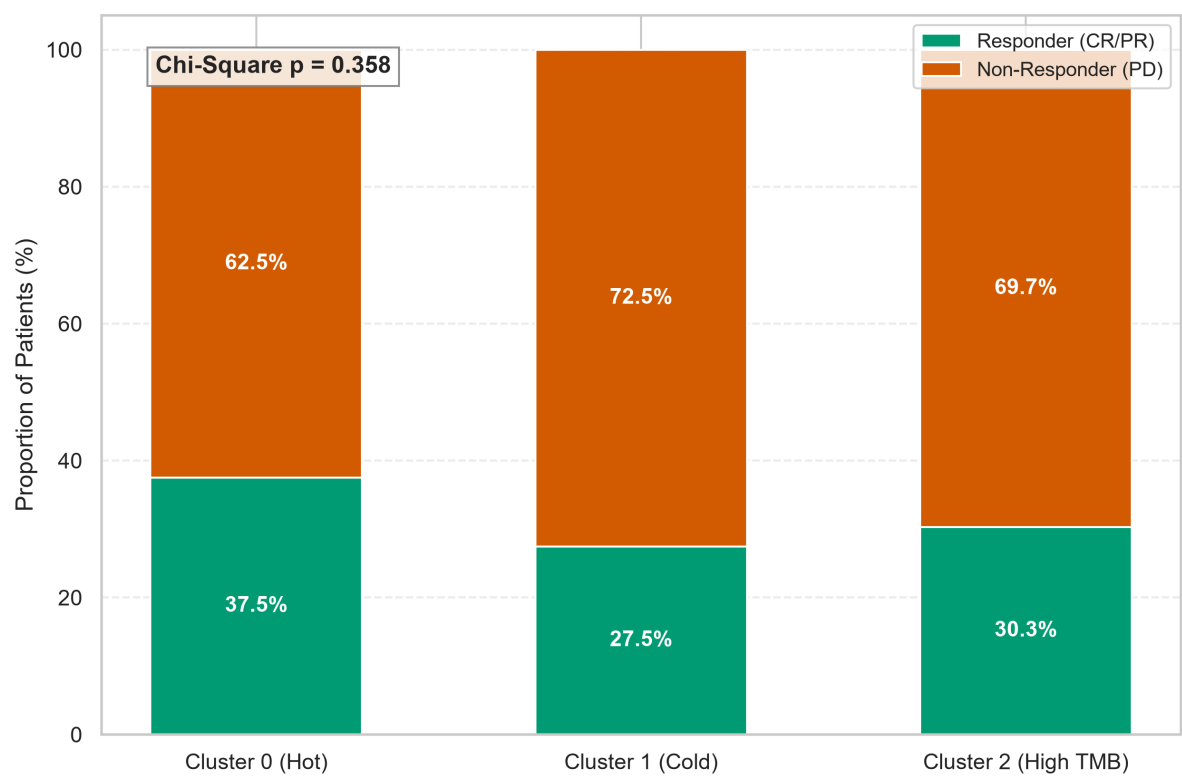


Immunotherapy Response & Overall Survival Validation

Validation across clinical outcomes demonstrates that unsupervised immune subtyping strongly correlates with clinical benefit:

- **Therapeutic Response Rate (Trial Cohorts, $N = 247$):** Significant difference in response rate across clusters (Chi-Square p-value = **(3.58e-01)**).

Immunotherapy Response Rate by Patient Subtype (Trial Cohorts, N=247)



- **Overall Survival (Full Dataset, $N = 699$):** Highly significant survival separation across patient subtypes (Log-Rank p -value = **(2.29e-05)**):

Full Dataset OS: Kaplan-Meier of Immunological Subtypes (N=677)

